SCIENCE · LIVING THINGS AND CLIMATE

Our Changing Climate

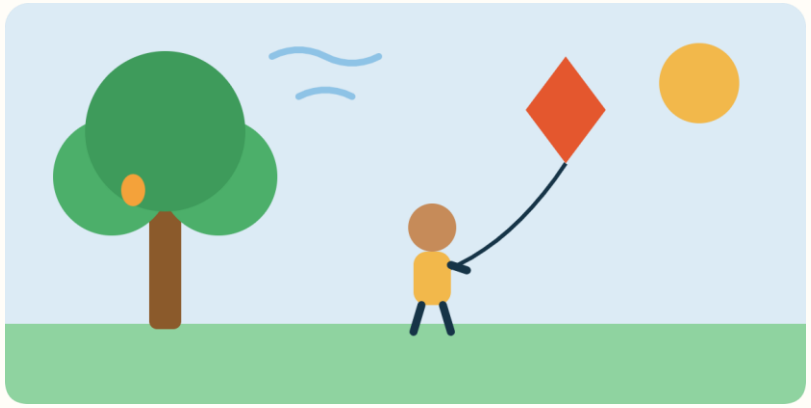
What climate change means, and how Philippine plants, animals, and people respond to it

Warmer air | Rising seas | Living things

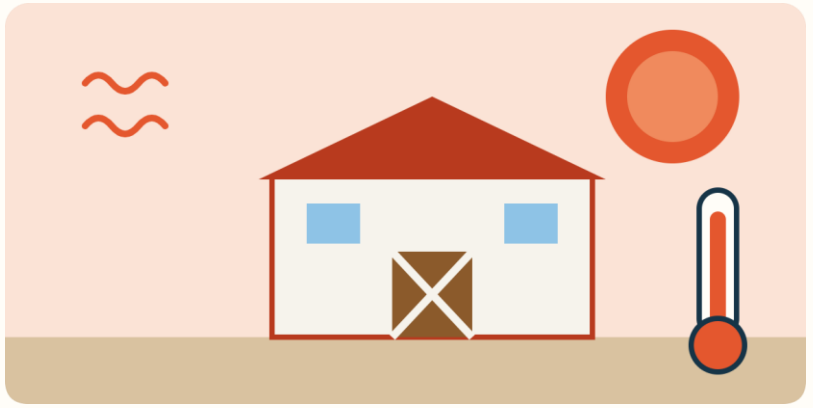


WARM-UP

Lola's summer vs. your summer



Lola's summer, long ago: "We played outside all afternoon. It was hot, but there was always a breeze."



Your summer, April 2025: the heat index hit **53°C** in Iba, Zambales, and many schools stopped in-person classes.

Just one hot summer, or something bigger?

LEARNING GOALS

Your mission today

1. Explain what **climate change** means and what causes it.

2. Describe how Philippine **plants, animals, and people respond** to climate change, and how they help scientists understand its effects.

Key words: weather · climate · greenhouse effect · coral bleaching · bioindicator · adaptation · mitigation

PART 1 · WHAT IS CLIMATE CHANGE?

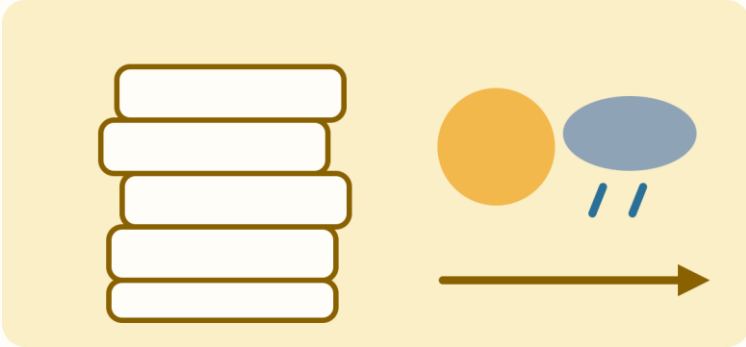
Weather vs. climate



Weather

The conditions **right now**: sunny, rainy, windy, or hot. It can change in a few hours.

Example: sunny this morning, rain at 3 p.m.



Climate

The **usual** weather pattern of a place over a long time, about 30 years.

Example: the Philippines is usually warm, with a wet and a dry season.

PART 1 · THE BIG DEFINITION

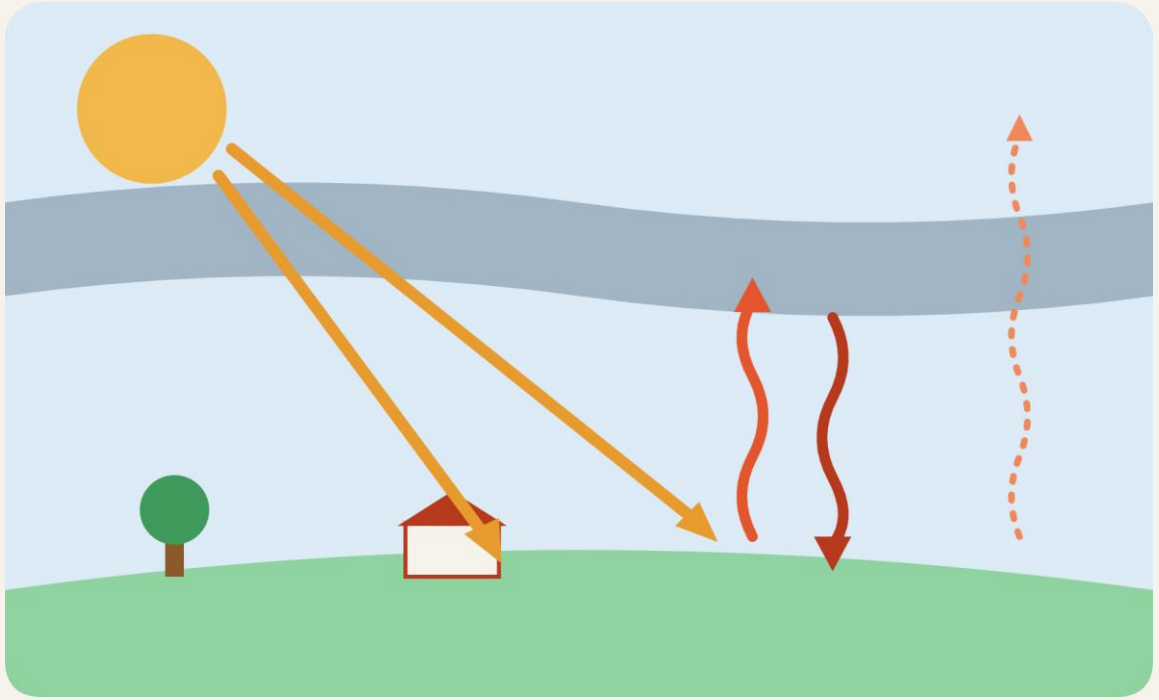
Climate change means long-term shifts in Earth's temperatures and weather patterns.

Since the 1800s, the main cause has been **human activities**, especially burning coal, oil, and gas.



PART 1 · HOW IT WORKS

The greenhouse effect



Yellow: sunlight in · **Orange:** heat bounced back · **Gray band:** greenhouse gases

Greenhouse gases, like carbon dioxide and methane, wrap Earth like a blanket.

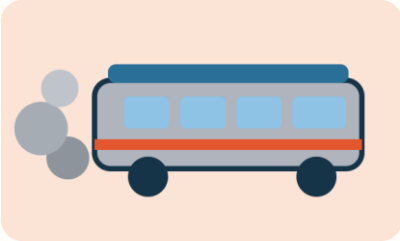
Sunlight gets in, but some heat can’t get out. **More gases = thicker blanket = warmer Earth.**



Like a jeepney parked in the sun with its windows shut: it keeps getting hotter inside.

PART 1 · CAUSES

Where do the extra gases come from?



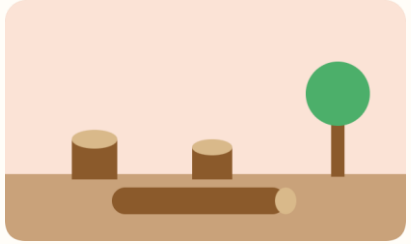
Vehicles

Jeepneys, cars, and buses burn gasoline and diesel.



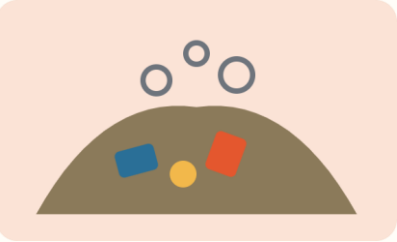
Power plants

Many burn coal to make our electricity.



Cutting forests

Fewer trees are left to soak up carbon dioxide.



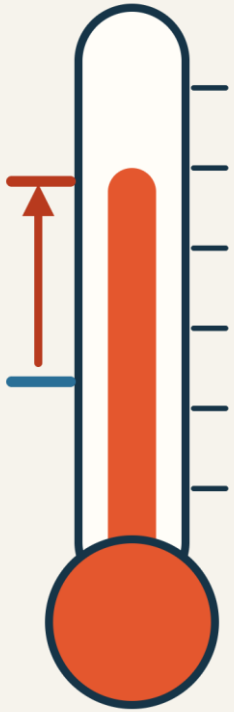
Garbage

Rotting trash in dumpsites gives off methane gas.

All of these add **extra greenhouse gases** to the air, making the blanket thicker.

PART 1 · EVIDENCE IN THE PHILIPPINES

Clue 1: the country is warming



+0.65°C

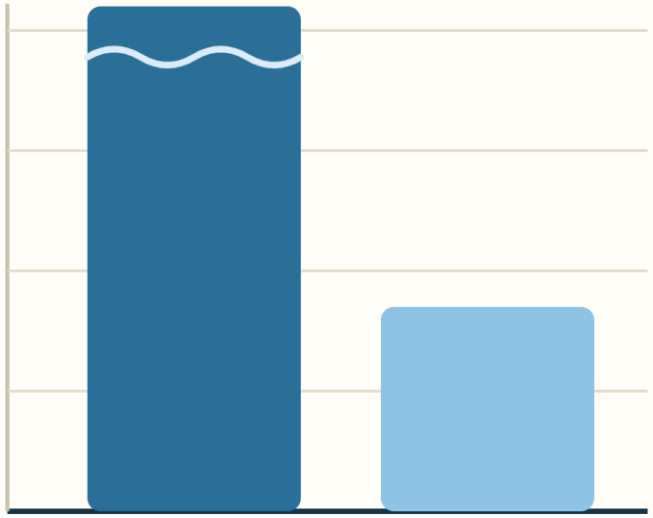
The Philippines' average yearly temperature rose by about **0.65°C** from 1951 to 2010.

Sounds small? Your body is normal at 37°C, but at 38°C you have a fever. Small rises matter!

Source: PAGASA temperature records, in the Philippine Climate Change Assessment report

PART 1 · EVIDENCE IN THE PHILIPPINES

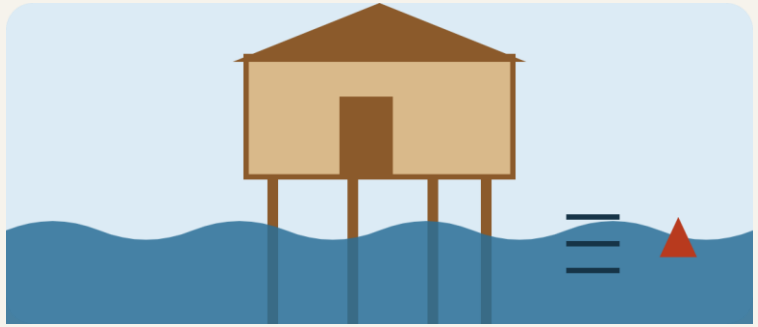
Clue 2: the sea is rising



Metro Manila
8.4 mm a year

World average
3.4 mm a year

From 1901 to 2022, Metro Manila's sea rose **almost 3 times faster** than the world average.

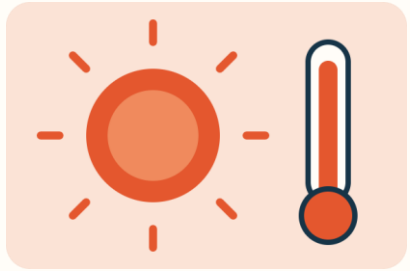


Warmer water expands, and melting ice adds more water. In some coastal towns, the land is also sinking.

Source: NAMRIA, reported by the Philippine Daily Inquirer

PART 1 · EVIDENCE IN THE PHILIPPINES

Clue 3: weather is getting wilder



Hotter days

April 2024: a **53°C** heat index in Iba, Zambales.



Heavier rain

More days with extreme rainfall in places like Tacloban and Iloilo.



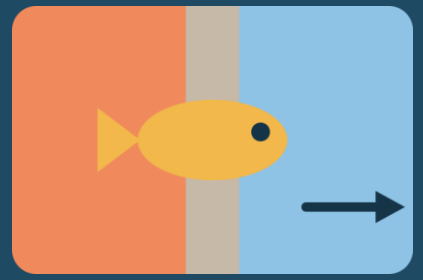
Stronger typhoons

Typhoons with winds of 150 km/h or more have become more frequent.

Sources: PAGASA (heat index, April 28, 2024); Philippine Climate Change Assessment report (rainfall and typhoons)

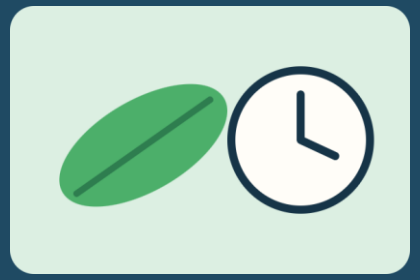
PART 2 · HOW LIVING THINGS RESPOND

Three ways to cope with the heat



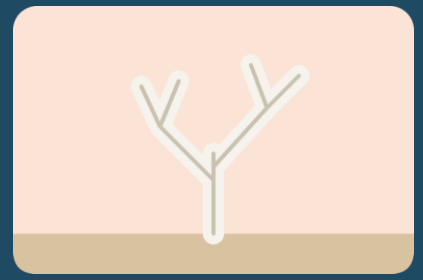
Move

Go somewhere cooler: deeper water or higher mountains.



Adjust

Change timing, behavior, or body: sweat, close pores, bloom earlier.

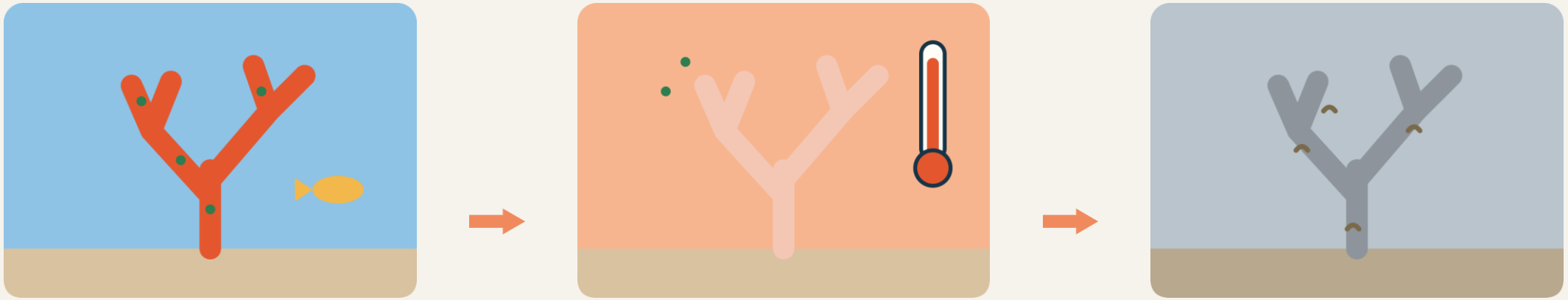


Struggle

If they can’t move or adjust, they weaken, get sick, or die.

PART 2 · EXAMPLE 1: CORAL REEFS · STRUGGLE

Why corals turn white



1 · Healthy

Tiny algae live inside corals, giving them food and color.

2 · Too hot

In hot water, corals push out the algae and turn white.

3 · Dead

If the heat lasts, the corals starve and die.

Real case, June 2024: divers at Escoda Shoal in the West Philippine Sea found almost all the corals they surveyed dead or dying after the water warmed.

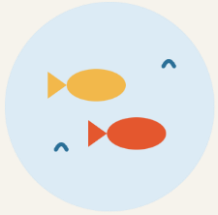
PART 2 · EXAMPLE 1: CORAL REEFS

The reef's ripple effect



Corals die

The reef "apartment building" crumbles.



Fish lose homes

Young fish have no place to hide and feed.



Fewer fish caught

Fishers bring home smaller catches.



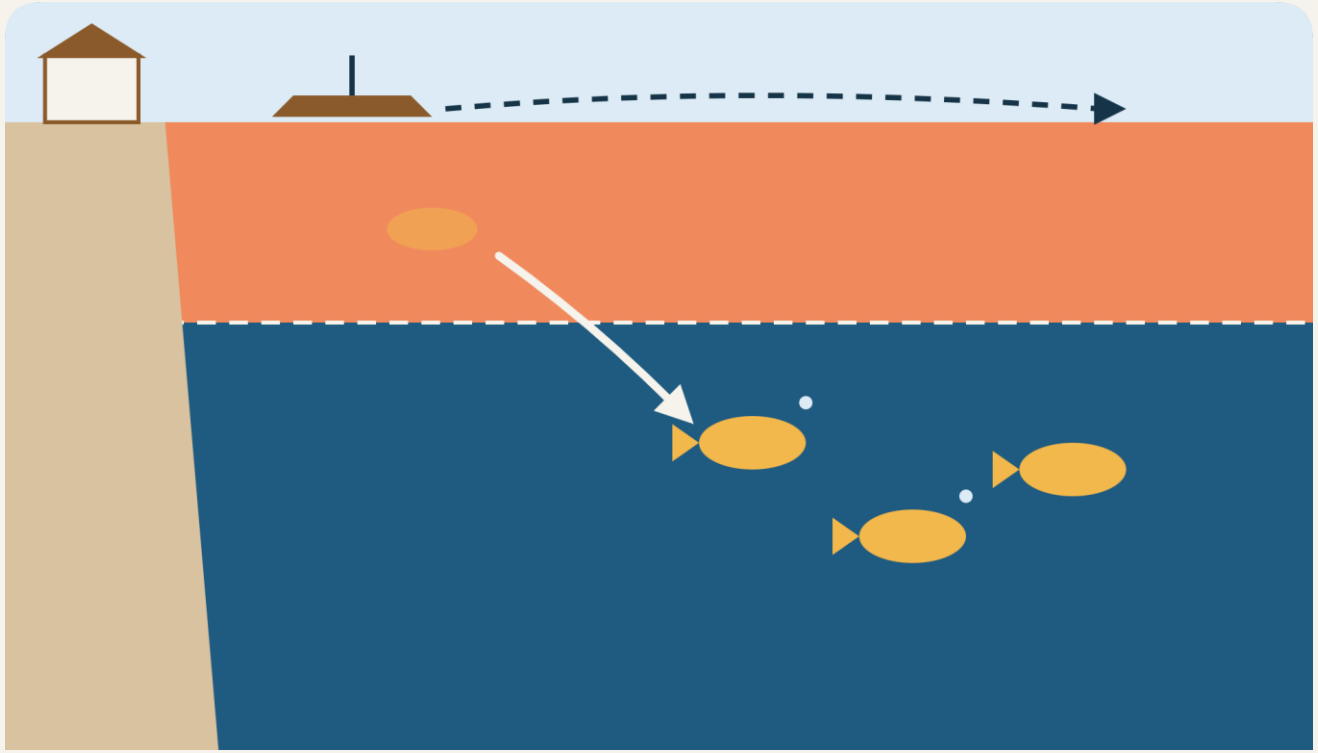
Families affected

Less food and less income for coastal towns.

Did you know? The Philippines is part of the Coral Triangle, the ocean's richest home for marine life.

PART 2 · EXAMPLE 2: FISH · MOVE

Fish swim away from the heat



Warm top layer: too hot for many fish

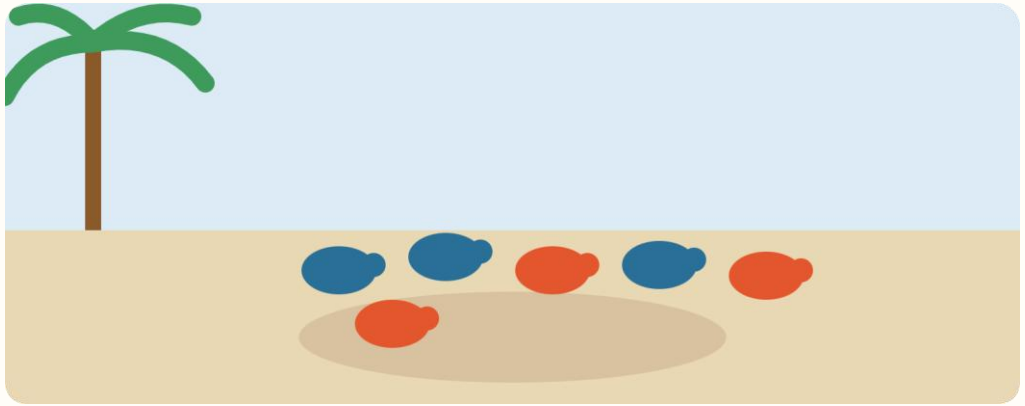
Cool deep water: where fish escape to

Many fish, like sardines (*tamban*), are sensitive to warm water. They move **deeper or farther** to stay cool.

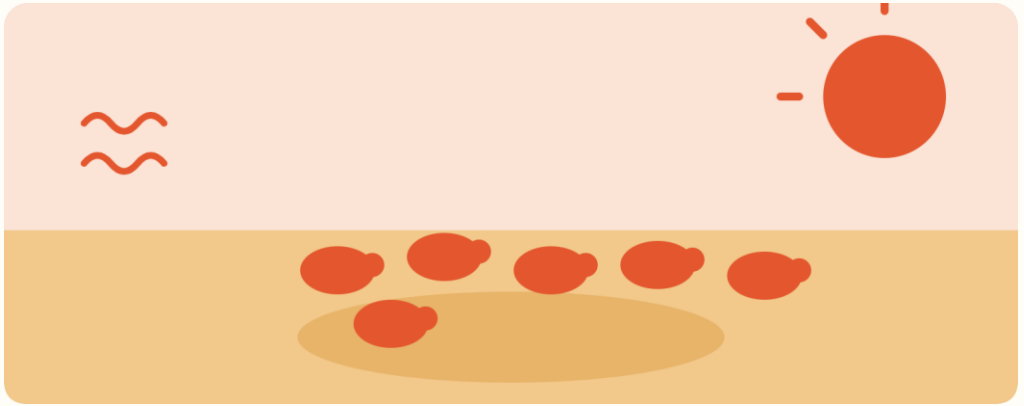
The effect on people: fishers must sail farther and use more fuel to catch them.

PART 2 · EXAMPLE 3: SEA TURTLES · STRUGGLE

Hot sand, more girl pawikan



Cooler sand: a healthy mix of **male (blue)** and **female (orange)** hatchlings.

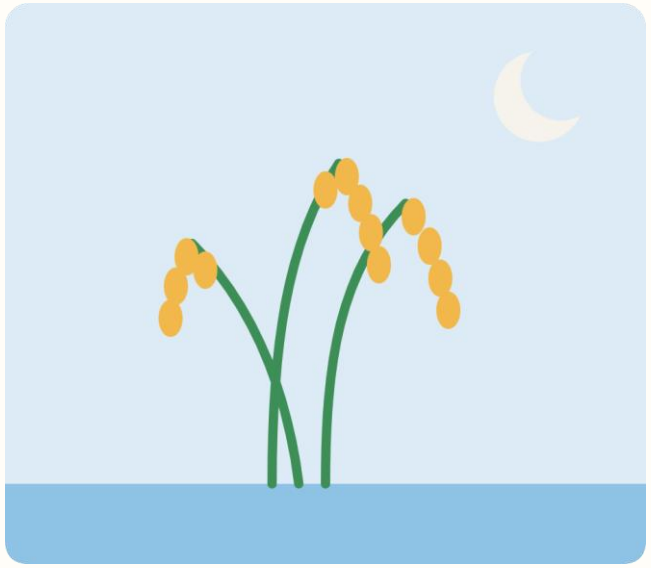


Hot sand: almost all the eggs hatch into **females**.

⚠ **Why it matters:** with too few males, fewer baby turtles can be born in the future. Rangers shade nests to help.

PART 2 · EXAMPLE 4: RICE · STRUGGLE

Hot nights, less rice



Cool nights: heavy heads full of grain



Hot nights: fewer, emptier grains

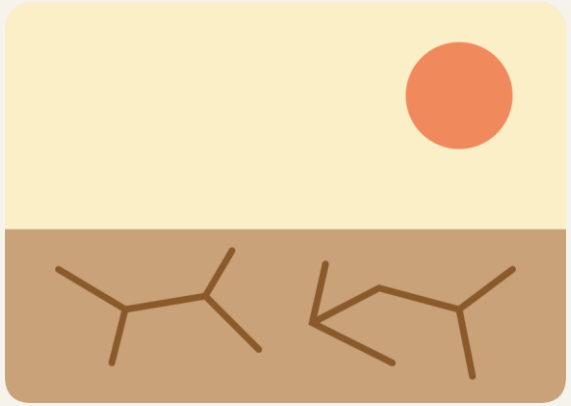
−10%

rice harvest for each **1°C** rise in nighttime temperature

IRRI study, Los Baños, Laguna (dry season, 1992–2003)

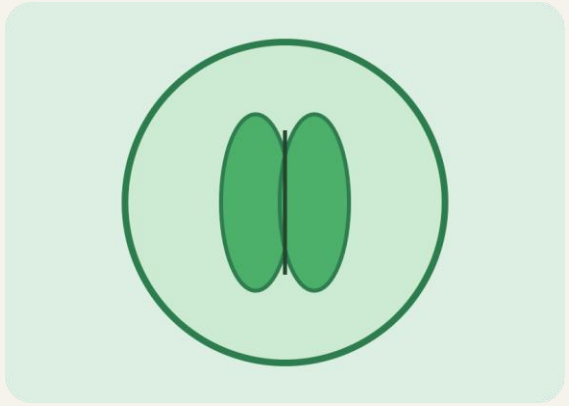
PART 2 · EXAMPLE 5: CROPS IN DROUGHT · ADJUST, THEN STRUGGLE

How a plant survives a dry spell



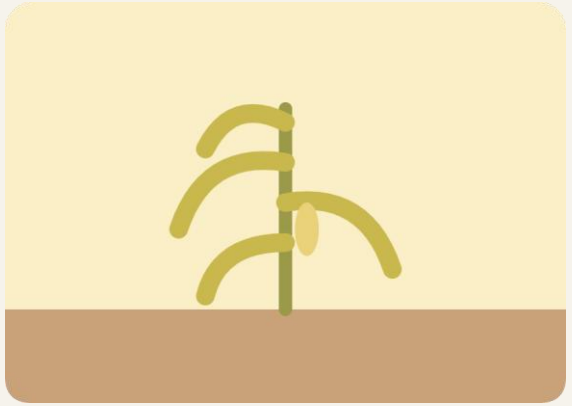
1 · Dry soil

El Niño brings less rain, so the soil dries out.



2 · Pores close

The plant **adjusts**: it shuts its stomata to save water.



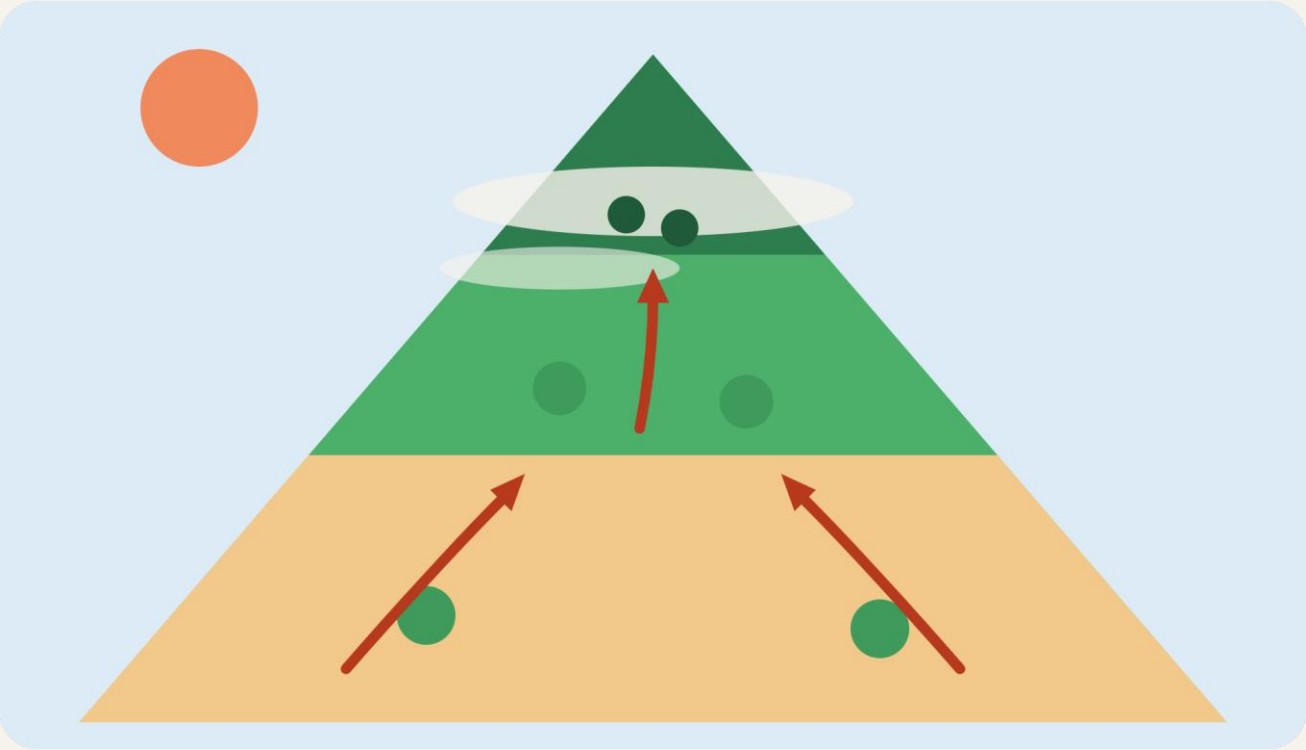
3 · Less food made

Less carbon dioxide gets in, so the plant **struggles**.

Remember photosynthesis? Closing the stomata saves water, but it also blocks the carbon dioxide leaves need to make food.

PART 2 · EXAMPLE 6: MOUNTAIN LIFE · MOVE

Climbing to stay cool



- **Cloud forest:** cool and misty
- **Middle forest**
- **Lowland:** getting warmer

As the lowlands warm, many plants and animals **move uphill** to cooler air.

The problem: the mountaintop is small. Species already at the top have nowhere left to go.

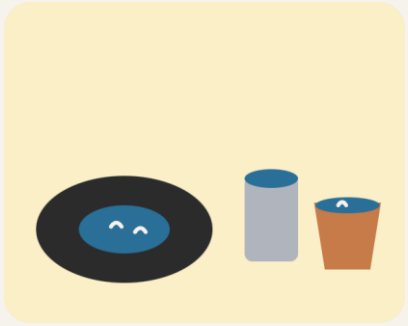
PART 2 · EXAMPLE 7: MOSQUITOES · THRIVE

Some living things love the change



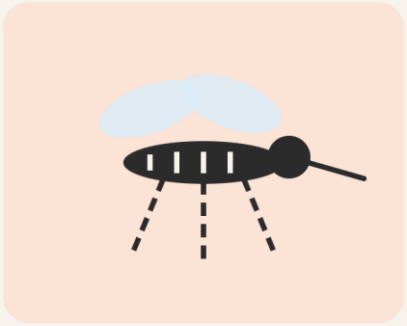
Warm and wet

Heavy rain, then heat



Standing water

More places for eggs to hatch



More mosquitoes

They grow faster in warm water

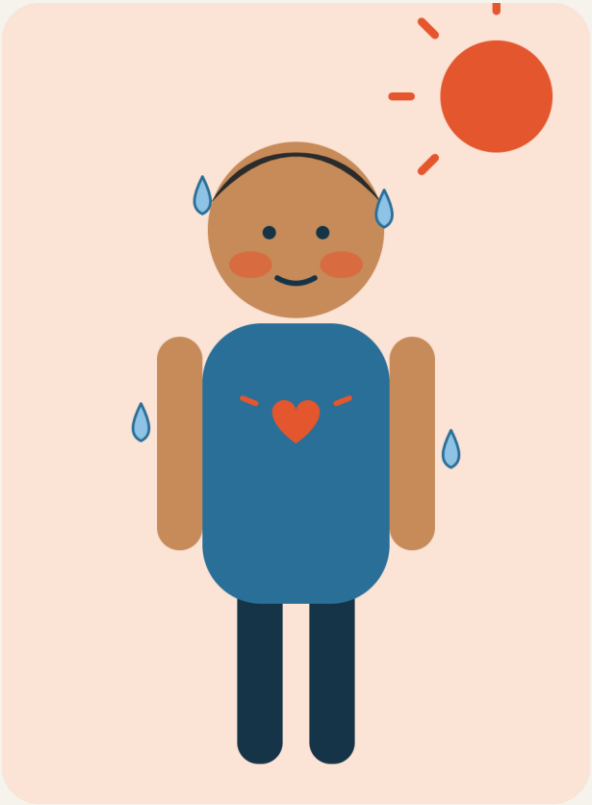


Dengue risk

More bites can spread disease

Your job: empty containers, cover water drums, and clean gutters every week.

Your body's cooling team

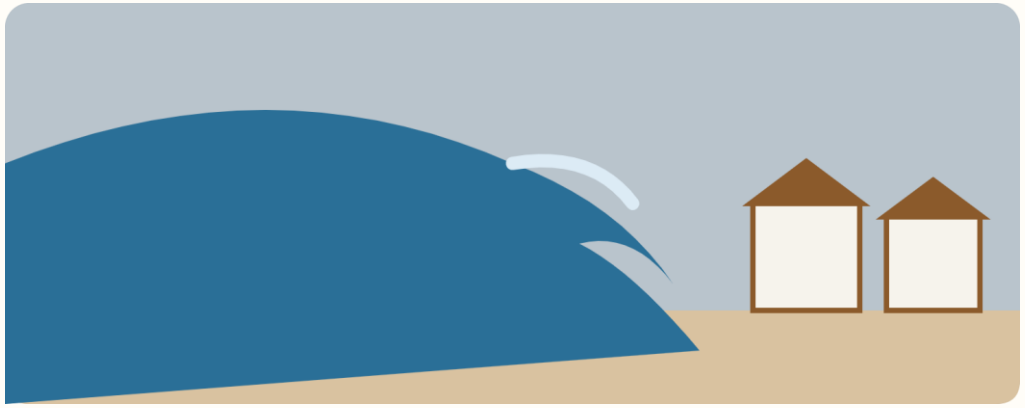


1. **Sweat:** as sweat evaporates, it carries heat away from your skin.
2. **Blood vessels widen:** more blood flows near the skin to release heat. That's why faces turn red.
3. **Heart beats faster:** it pumps more blood to the skin.

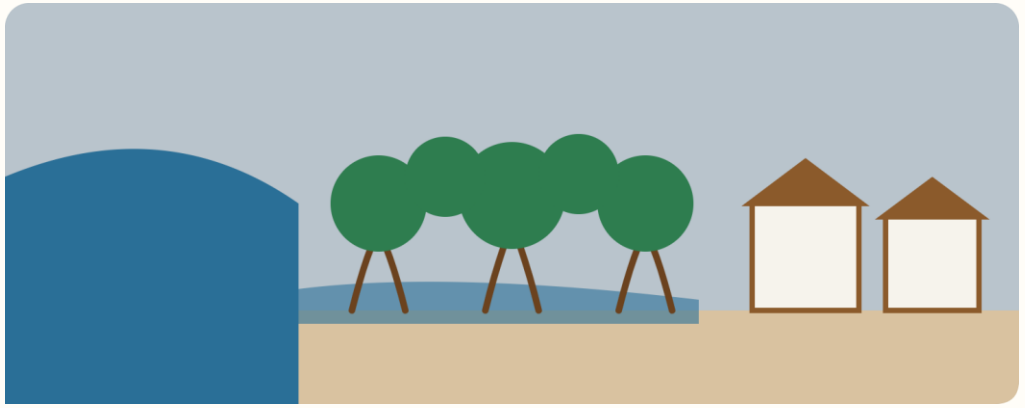
⚠ **Too much heat** overwhelms the team: dizziness, cramps, even heat stroke. Drink water, find shade, and rest.

PART 2 · EXAMPLE 9: MANGROVES · NATURE'S HELPERS

Mangroves: our coastal shields



No mangroves: big waves hit homes at full force.



With mangroves: roots and trunks slow the wave down.

Up to 66% smaller waves behind 100 m of mangroves

613,000 Filipinos spared from floods every year

Carbon stores: roots and mud lock away carbon

PART 3 · LEARNING FROM LIVING THINGS

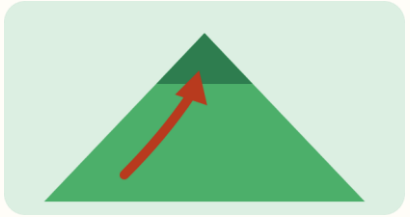
Nature’s climate detectives

A **bioindicator** is a living thing whose health and behavior tell us how the environment is changing.



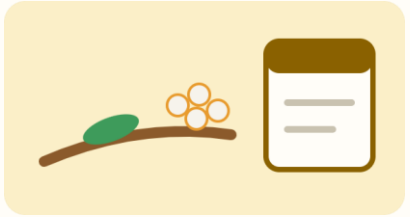
Bleached corals

Tell us: the sea is getting too warm.



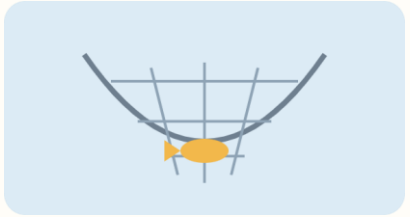
Species moving uphill

Tell us: the lowlands are heating up.



Flowering out of season

Tell us: the usual seasons are shifting.



Smaller fish catches

Tell us: ocean life is changing.

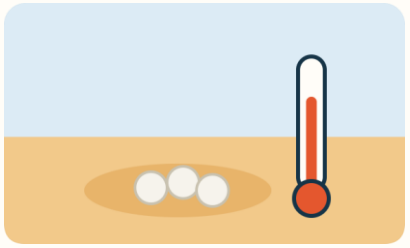
PART 3 · LEARNING FROM LIVING THINGS

How scientists read the clues



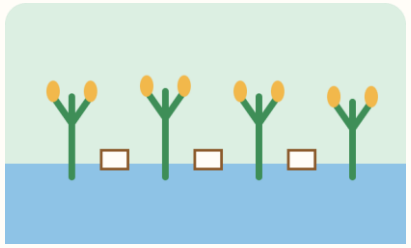
Reef surveys

Divers count healthy and bleached corals.



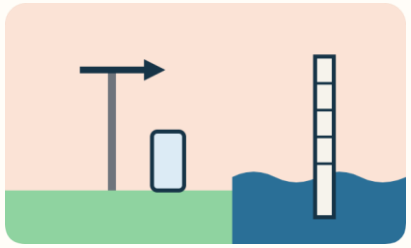
Nest watch

Rangers record turtle nests and sand temperatures.



Rice trials

IRRI tests rice in heat and breeds tougher types.



Long-term records

PAGASA and NAMRIA track heat, rain, and sea level.

Living clues + long-term records = a clearer picture of how climate change affects Filipinos.

PART 3 · WHAT WE CAN DO

Slow it down and get ready

Mitigation

Slow climate change by cutting greenhouse gases.

- Turn off lights and fans you’re not using.
- Walk, bike, or share rides when you can.
- Plant trees, and reduce, reuse, recycle.

Adaptation

Get ready by protecting people and living things.

- Plant and protect mangroves along coasts.
- Grow crops that can handle heat and drought.
- Drink water on hot days, and clear mosquito breeding spots.

SUMMARY · THE BIG PICTURE

The whole story in six steps



1 · People

burn fuels and cut forests



2 · Gases

thicken Earth's heat blanket



3 · Climate

gets hotter, wetter, and wilder



4 · Life

moves, adjusts, or struggles



5 · Scientists

read the living clues



6 · We act

slow it down and get ready

Every living thing feels the change, and every one of us can help.

TAKE-HOME MISSION

Be a climate detective!

1 · Track the heat

For one week, note the day's heat index from a PAGASA report.

2 · Spot a living clue

Watch a plant, insect, or animal near home. How is it coping with the weather?

3 · Draw and share

Sketch what you saw and write what it might tell us about our climate.

Small actions, big difference. Let's care for our home!